BSC STATISTICS — FINAL YEAR PROJECT

Statistical Modelling for *F1 Driver* Progression →

Using FIA Formula 2 and FIA Formula 3 Performance Data

Presented by: Chris Joemon | Supervisor: Prof. M. Syluvai Anthony | Academic Year 2025–2026

INTRODUCTION

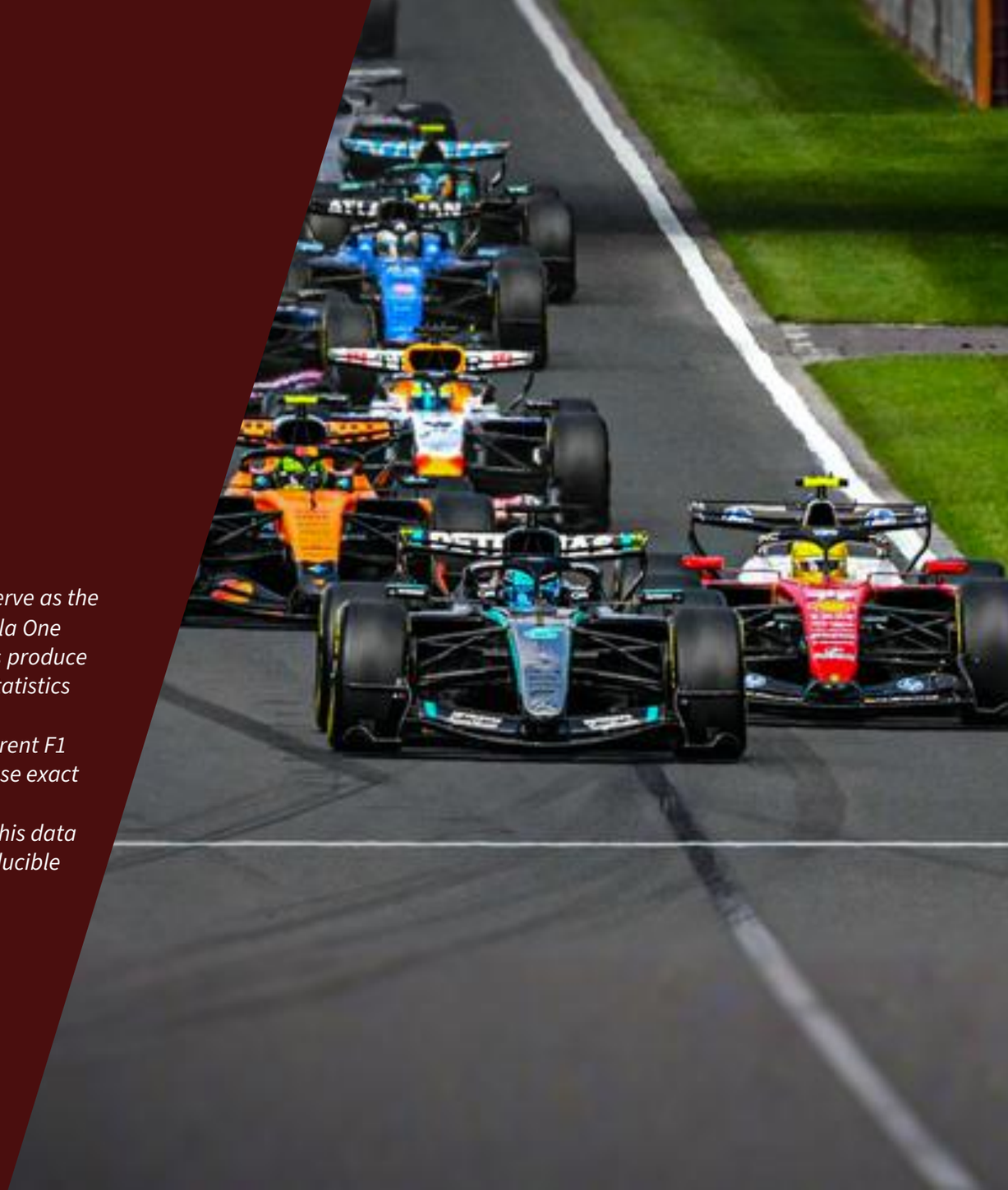
Project Background & Motivation

The Problem

Formula One team principals and talent scouts face significant uncertainty when identifying which junior drivers will successfully transition to F1. Current scouting relies heavily on subjective judgment, with limited data-driven tools to support decision-making.

Why F2 & F3 Data?

- *FIA Formula 2 and Formula 3 serve as the primary feeder series to Formula One*
- *Drivers in these championships produce rich, structured performance statistics season-by-season*
- *Historically, the majority of current F1 drivers progressed through these exact series*
- *A statistical model trained on this data could provide objective, reproducible predictions*



OBJECTIVES

Objectives of the Study

Build Predictive Models

Develop and compare machine learning classification models to predict the likelihood of a junior driver progressing to Formula One based on their F2/F3 performance metrics.

Rank Driver Potential

Generate a ranked list of current and recent F2/F3 drivers by their predicted F1 progression probability, providing a data-driven talent identification framework.

Construct a Suitability Index

Create a Driver Suitability Index (DSI) that consolidates key performance variables into a single composite score to assess F1 readiness in a transparent, interpretable way.

DATA

Dataset Description

453

Observations

Driver-season records spanning 2019–2025

2

Series Covered

FIA Formula 2 and FIA Formula 3 Championships

7

Seasons

2019, 2020, 2021, 2022, 2023, 2024, 2025

1

Binary Outcome

Progressed to F1 (Yes = 1 / No = 0)

Key Variables Collected

Performance

- *Points per race (average)*
- *Final championship rank*
- *Pole positions per season*
- *Fastest laps per season*

Reliability & Profile

- *DNF (Did Not Finish) rate*
- *Podium finish rate*
- *Driver age at season start*
- *Series level (F2 or F3)*

Data sourced from official FIA results archives and publicly available championship statistics. Missing values were handled via median imputation for continuous variables.

DATA

Dataset

driver_name	season	series	total_points	wins	podiums	final_rank	age	f1_progression	f1_entry_year	pole_position	total_races	dnfs	points_per_race
N. de Vries	2019	F2	266.0	4	12	1	24	1	2023.0	5	22	0	12.091
N. Latifi	2019	F2	214.0	4	8	2	24	0	NaN	0	22	0	9.727
L. Ghiotto	2019	F2	207.0	4	9	3	24	0	NaN	2	22	3	9.409
S. Sette Câmara	2019	F2	204.0	2	8	4	21	0	NaN	2	22	2	9.273
J. Aitken	2019	F2	159.0	3	7	5	24	0	NaN	0	22	1	7.227
...	...	...	...	...	...	...	...	...	...	...	...	...	...
N. Johnson	2025	F3	0.0	0	0	33	17	0	NaN	0	16	4	0.000
J. Sagrera	2025	F3	0.0	0	0	34	21	0	NaN	0	14	3	0.000
F. Barrichello	2025	F3	0.0	0	0	35	20	0	NaN	0	6	1	0.000
N. Marinangeli	2025	F3	0.0	0	0	36	21	0	NaN	0	4	2	0.000
J. Garfias	2025	F3	0.0	0	0	37	21	0	NaN	0	4	2	0.000

EDA

Exploratory Data Analysis & *Key Insights*

Class Imbalance Identified

*Approximately **18% of drivers** in the dataset progressed to F1, reflecting the highly competitive and selective nature of the pathway. SMOTE oversampling was applied to address the imbalance during model training.*

Points Per Race is the Strongest Predictor

*Drivers who progressed to F1 recorded a **significantly higher average points per race** compared to those who did not. The distribution of this variable showed clear separation between the two outcome groups in violin and box plots.*

DNF Rate Negatively Correlates with Progression

A higher DNF rate was consistently associated with non-progression. Reliability appears to signal driver maturity and racecraft — qualities F1 teams prioritize. Final rank and pole positions also showed strong positive correlation with progression.

DATA

EDA

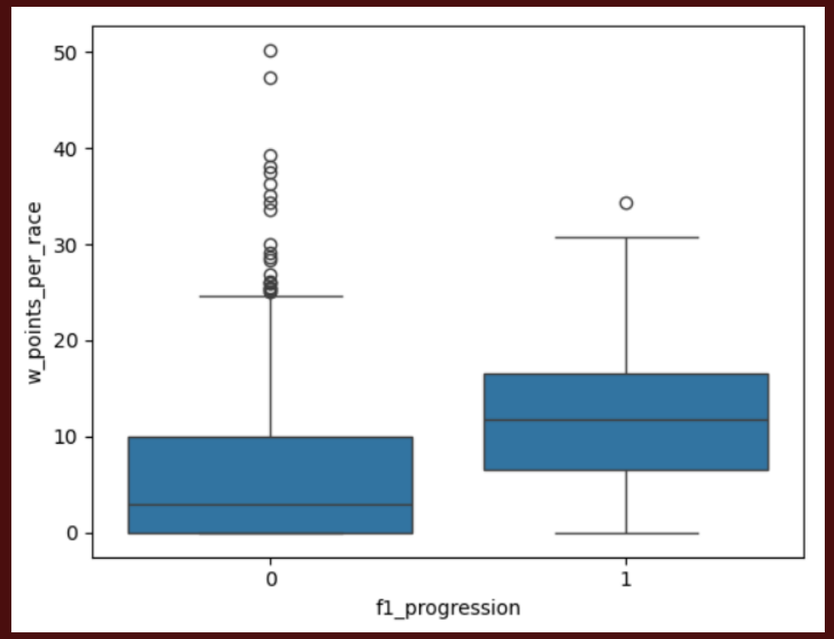
f1_progression	s.no	season	total_points	wins	podiums	final_rank	age
0	233.501205	2022.168675	45.622892	0.551807	1.715663	17.857831	20.640964
1	156.000000	2021.210526	120.644737	1.947368	4.894737	7.921053	19.552632

f1_entry_year	pole_position	total_races	dnfs	points_per_race	win_rate	podium_rate	dnf_rate	season_weight	w_points_per_race	w_pole_position
NaN	0.257831	16.93012	2.067470	2.214133	0.025952	0.082292	0.122795	3.081089	6.638662	0.783771
2023.552632	1.184211	20.50000	2.342105	5.491632	0.086842	0.222579	0.121237	2.188389	11.936901	2.636790

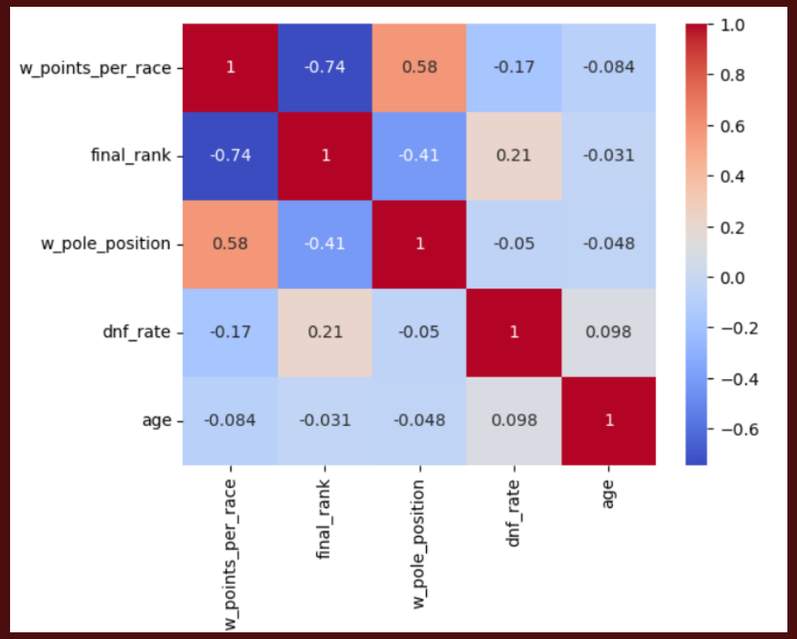
Descriptive Statistics w.r.t. the target variables

DATA

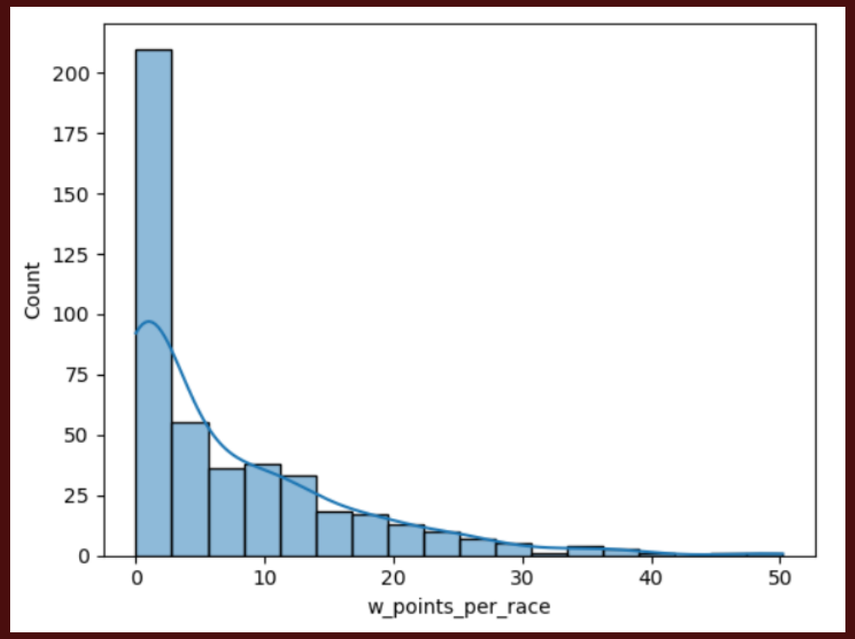
EDA



Box plot



Correlation Matrix



Histogram

HYPOTHESIS TESTING

Hypothesis Testing & *Significant Variables*

Tests Conducted

Independent Samples t-Test

Compared mean points per race and podium rates between drivers who progressed vs. those who did not. Results confirmed statistically significant differences ($p < 0.001$).

Chi-Square Test

Applied to categorical variables such as series level (F2 vs. F3). F2 participation was found to be significantly associated with F1 progression ($p < 0.05$).

Correlation Analysis

Pearson and Spearman correlations identified multicollinearity concerns between rank and points-related variables. VIF analysis informed feature selection for the regression model.

Statistically Significant Predictors ($p < 0.05$)

✅ Points Per Race

Strongest positive predictor across all tests

✅ Final Championship Rank

Top-3 finishes highly predictive of progression

✅ Pole Positions

Positive association with F1 readiness

✅ DNF Rate

Significant negative predictor

Age was found to be marginally significant ($p = 0.048$), with younger drivers showing slightly higher progression rates at equal performance levels.

METHODOLOGY

Predictive Models Used

Logistic Regression

- *Baseline classification model for binary outcome (progressed / not progressed)*
- *Provides interpretable odds ratios for each predictor variable*
- *L2 regularisation applied to handle multicollinearity*
- *10-fold cross-validation used for robust evaluation*

Random Forest

- *Ensemble of 500 decision trees trained via bagging*
- *Captures non-linear relationships and interaction effects*
- *Feature importance scores extracted for variable interpretation*
- *Hyperparameter tuning via grid search (max depth, n_estimators)*

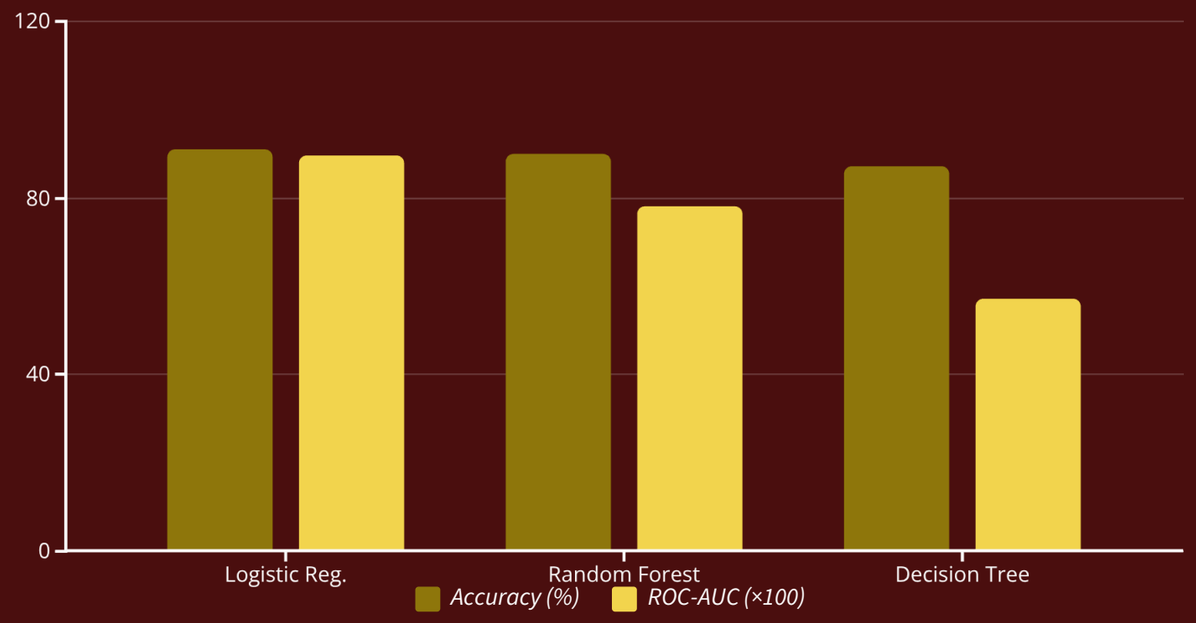
Decision Tree

- *Single classification tree with Gini impurity splitting criterion*
- *Highly interpretable — visual tree can be shown to non-technical audiences*
- *Maximum depth constrained to prevent overfitting*
- *Serves as a transparent benchmark for the ensemble methods*

All models were trained on an 80/20 train-test split following SMOTE oversampling on the training set only, with stratification to preserve class proportions.

RESULTS

Model Comparison *Results*



Performance Summary

🥇 Logistic Regression — Best Overall

Accuracy: 91% | ROC-AUC: 0.897*Despite its simplicity, logistic regression achieved the highest discrimination ability. Strong ROC-AUC indicates excellent separation between progressors and non-progressors.*

🥈 Random Forest

Accuracy: 90% | ROC-AUC: 0.78*Marginally lower accuracy than logistic regression. Strong for capturing non-linear patterns; provided most informative feature importance rankings.*

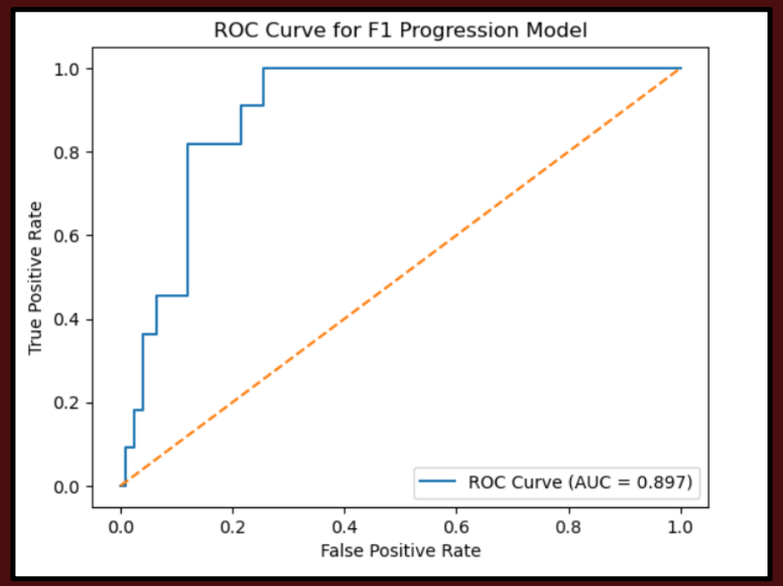
🥉 Decision Tree

Accuracy: 87% | ROC-AUC: 0.57*Lowest discriminatory power despite reasonable accuracy. Prone to overfitting; low AUC suggests limited ability to differentiate borderline cases.*

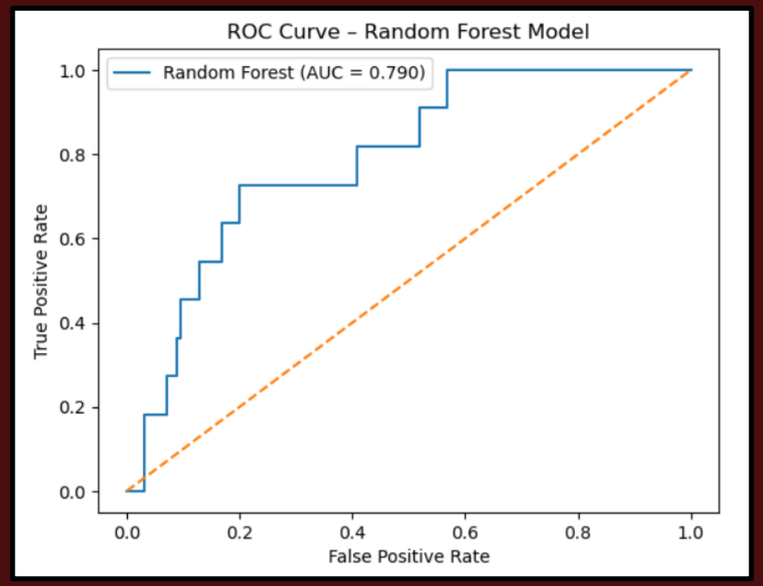
ROC-AUC is the primary evaluation metric here as the dataset is class-imbalanced. Accuracy alone can be misleading with imbalanced classes.

RESULTS

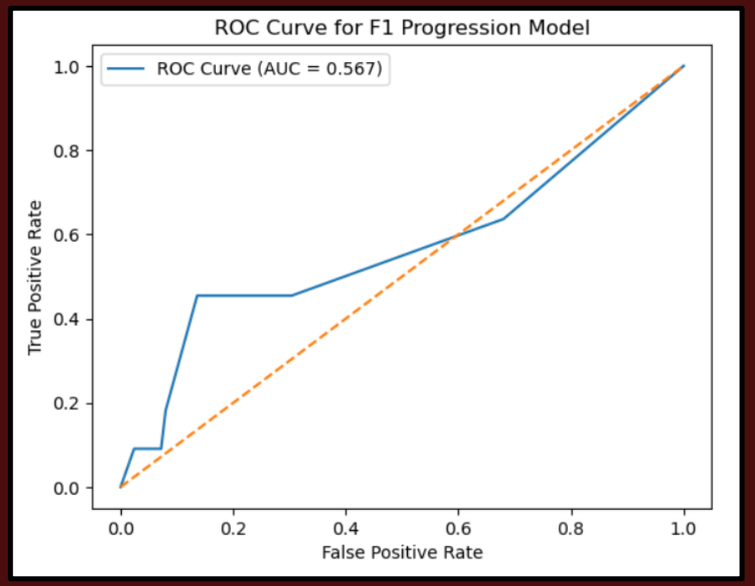
ROC Curve



Logistic Regression



Random Forest



Decision Tree

OUTPUTS

Driver Ranking & *Suitability Index*

Driver Ranking by Predicted Probability

Using the logistic regression model's output probabilities, all 453 driver-season observations were ranked from highest to lowest predicted likelihood of F1 progression. This enables direct comparison across seasons and series, surfacing the most statistically promising talent.

Driver Suitability Index (DSI)

A composite scoring index was constructed using the five most significant predictors — weighted by their logistic regression coefficients. The DSI condenses multi-dimensional performance into a single interpretable score between 0 and 100 for clear benchmarking.

DSI Selection Conditions — A Driver is Flagged as "F1-Ready" if:

- Points Per Race ≥ 8.0
- Final Rank ≤ 5th
- DNF Rate ≤ 15%
- Predicted Prob. ≥ 0.60
- Age ≤ 23 Years

OUTPUTS

Conditions used to Rank the Driver

Condition	Reason
High Points per Race	Indicates consistent race performance and scoring ability
Strong Final Championship Rank (Top positions)	Reflects competitiveness throughout the season
High Number of Pole Positions	Indicates strong qualifying pace and raw speed
Low DNF Rate	Represents reliability and race completion consistency
Younger Age	Indicates long-term development potential for teams
High Predicted Probability from Model	Shows statistical likelihood of progressing to F1

OUTPUTS

Final Driver Rankings

	driver_name	season	predicted_probability	driver_suitability_index
416	R. Câmara	2025	0.902916	4.245779
317	P. Aron	2024	0.602756	2.055619
216	Z. Maloney	2022	0.458949	1.171131
286	G. Minì	2023	0.421224	1.149743
151	D. Hauger	2021	0.421102	1.415028
62	C. Ilott	2020	0.418280	1.036344
90	F. Vesti	2020	0.400300	0.623346
259	V. Martins	2023	0.391673	1.392192
126	T. Pourchaire	2021	0.390272	0.626482
284	J. Martí	2023	0.382745	1.347780

Top 10 Driver Rankings

	driver_name	season	predicted_probability	driver_suitability_index
416	R. Câmara	2025	0.902916	4.245779
317	P. Aron	2024	0.602756	2.055619

Best two Drivers compatible for F1

CONCLUSION

Key Findings, Conclusion & *Future Scope*

Key Findings

Logistic Regression Outperforms

Achieved 91% accuracy and ROC-AUC of 0.897 — the best performing model for this binary classification problem, suggesting linear separability in the feature space.

Points Per Race is Most Predictive

Consistent across all three models as the single most important variable, followed by final championship rank and DNF rate.

DSI Provides Actionable Insight

The Driver Suitability Index translates complex model outputs into a practical, interpretable tool suitable for use by scouts and team analysts.

Conclusion & Future Scope

*This study demonstrates that **statistical modelling can reliably predict F1 progression** from junior formula performance data with high accuracy. The logistic regression model is recommended as the primary predictive tool due to its superior ROC-AUC and interpretability.*

Future Work

1 Expand Dataset

Include F1 performance data post-progression to validate model quality longitudinally

2 Add Team Context

Incorporate car quality and team funding as covariates to adjust for machinery advantage

3 Neural Networks

Explore deep learning approaches including LSTM for sequential season-by-season progression modelling

BSC STATISTICS — FINAL YEAR PROJECT

Statistical Modelling for *F1 Driver* Progression

Using FIA Formula 2 and FIA Formula 3 Performance Data

Presented by: Chris Joemon | Supervisor: Prof. M. Syluvai Anthony | Academic Year 2025–2026